

## AHA SCIENTIFIC STATEMENT

# Cardiopulmonary Impact of Electronic Cigarettes and Vaping Products: A Scientific Statement From the American Heart Association

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**ABSTRACT:** Vaping and electronic cigarette (e-cigarette) use have grown exponentially in the past decade, particularly among youth and young adults. Cigarette smoking is a risk factor for both cardiovascular and pulmonary disease. Because of their more limited ingredients and the absence of combustion, e-cigarettes and vaping products are often touted as safer alternative and potential tobacco-cessation products. The outbreak of e-cigarette or vaping product use–associated lung injury in the United States in 2019, which led to >2800 hospitalizations, highlighted the risks of e-cigarettes and vaping products. Currently, all e-cigarettes are regulated as tobacco products and thus do not undergo the premarket animal and human safety studies required of a drug product or medical device. Because youth prevalence of e-cigarette and vaping product use was as high as 27.5% in high school students in 2019 in the United States, it is critical to assess the short-term and long-term health effects of these products, as well as the development of interventional and public health efforts to reduce youth use. The objectives of this scientific statement are (1) to describe and discuss e-cigarettes and vaping products use patterns among youth and adults; (2) to identify harmful and potentially harmful constituents in vaping aerosols; (3) to critically assess the molecular, animal, and clinical evidence on the acute and chronic cardiovascular and pulmonary risks of e-cigarette and vaping products use; (4) to describe the current evidence of e-cigarettes and vaping products as potential tobacco-cessation products; and (5) to summarize current public health and regulatory efforts of e-cigarettes and vaping products. It is timely, therefore, to review the short-term and especially the long-term implications of e-cigarettes and vaping products on cardiopulmonary health. Early molecular and clinical evidence suggests various acute physiological effects from electronic nicotine delivery systems, particularly those containing nicotine. Additional clinical and animal-exposure model research is critically needed as the use of these products continues to grow.

**Key Words:** AHA Scientific Statements ■ adolescent ■ electronic cigarettes ■ electronic nicotine delivery systems ■ heart ■ lung ■ tobacco products ■ vaping

**E**lectronic (e-) cigarettes were created in the early 2000s in China by pharmacist Hon Lik as a device that generates nicotine vapor without smoke.<sup>1</sup> The basic mechanism behind the technology is the heating or atomization of a liquid solution, most commonly containing nicotine, stored in a reservoir (Supplemental Figure 1).<sup>1,2</sup> The aerosol generated from an e-cigarette or vaping product is inhaled. The solution, or e-liquid, generally contains a humectant, nicotine, and flavoring agents.<sup>3</sup> E-liquid

formulations contain mostly nicotine but can also contain other drugs, most commonly tetrahydrocannabinol but also other substances such as methamphetamine, methadone, and vitamins.<sup>4</sup> In 2007, e-cigarette products began to be introduced into the United States. The US Food and Drug Administration (FDA) attempted to halt the importation of these products under the assumption these were drug-device combination products.<sup>5</sup> After years of litigation, a 2010 court ruling, *Smoking Everywhere*,

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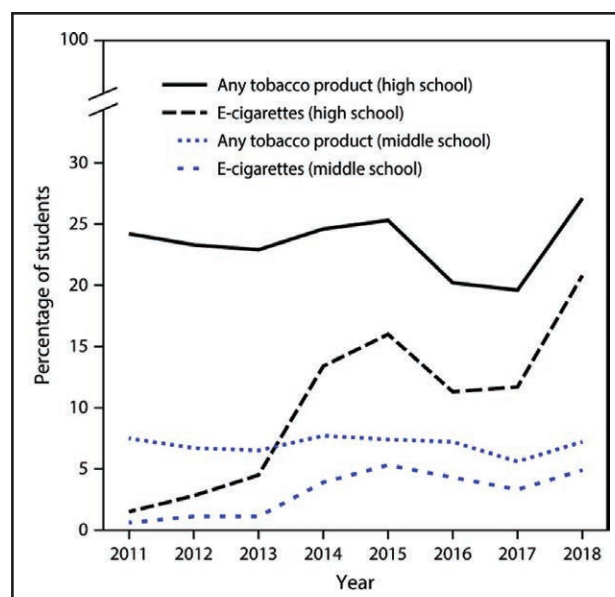
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Inc. vs US Food and Drug Administration, deemed e-cigarettes should be considered tobacco products, and thus, if the FDA were to regulate them, they should fall under the rubric of the recently enacted 2009 Family Smoking Prevention and Tobacco Control Act.<sup>5–7</sup> Thus began the adoption of e-cigarettes and vaping products into the US consumer market as tobacco products.<sup>5</sup> Since their introduction, different designs of electronic nicotine delivery system (ENDS) have been developed, with newer generations designed to withstand higher voltage levels without resulting in an unpleasant taste associated with overheating the flavored liquid that could result in higher reactive free radical concentrations.<sup>1,8</sup> The American Heart Association has published policy statements on e-cigarettes.<sup>9</sup> Accordingly, the objectives of this scientific statement are (1) to describe and discuss e-cigarettes and vaping products use patterns among youth and adults; (2) to identify harmful and potentially harmful constituents in vaping aerosols; (3) to critically assess the molecular, animal, and clinical evidence on the short- and long-term cardiovascular and pulmonary risks of e-cigarette and vaping products use; (4) to describe the existing evidence on the utility of e-cigarettes and vaping products as tobacco-cessation products; and (5) to summarize current public health and regulatory efforts of e-cigarettes and vaping products.

## EPIDEMIOLOGY AND HISTORY OF INCREASING INCIDENCE OF VAPING AND ELECTRONIC CIGARETTE USE

### ENDS Product Use in the Young

In the United States, ENDS products are the most commonly used tobacco product among youth, with a growing percentage of users who do not report ever smoking combustible cigarettes.<sup>10–12</sup> Data from the 2011 to 2018 NYTS (National Youth Tobacco Survey) showed an exponential increase in ENDS use among high school students; current (past 30-day) use increased from 1.5% in 2011 to 20.8% in 2018, an estimated 3.1 million students (Figure 1).<sup>11,13</sup> Among middle school students, current e-cigarette use increased from 0.6% in 2011 to 4.9% in 2018, an estimated 570 000 students.<sup>11,13</sup> Data from the 2019 NYTS showed that 27.5% of high school students (4.1 million) reported current e-cigarette use, whereas 5.8% reported using combustible cigarettes.<sup>10</sup> Data from Monitoring the Future Survey showed that current e-cigarette use more than doubled from 2017 to 2019 among middle and high school students.<sup>14</sup> In 2019, 25.5% of 12th graders reported current e-cigarette use compared with 11.0% in 2017. Current tetrahydrocannabinol vaping also dramatically increased from 2017 to 2019 in all grades, from 4.9% to 14.0% of 12th graders, 4.3% to 12.6% of 10th graders, and 1.6% to 3.9% of eighth graders. An estimated 72.2% of current exclusive e-cigarette



**Figure 1. Percentage of middle and high school students who currently use electronic cigarettes and any tobacco product, National Youth Tobacco Survey, United States, 2011 to 2018.<sup>11</sup>**

users reported use of flavored products.<sup>10</sup> Among middle school students, 10.5% (1.2 million) reported current e-cigarette use compared with 2.3% who reported combustible cigarette smoking. Among middle school students who reported exclusive ENDS use, 59.2% reported using flavors, with fruits, menthol or mint, and candy, desserts, or other sweets as the most commonly used flavors.<sup>10</sup> These data are consistent with findings from other studies indicating the rapid increase in prevalence of e-cigarette use among youth in other countries such as Canada and England.<sup>15,16</sup> The prevalence of ENDS use among youth remained stable despite the emergence of the coronavirus disease 2019 (COVID-19) pandemic and growing concern about the increased risks of COVID-19 from tobacco and ENDS products.<sup>17,18</sup> Data from the 2020 NYTS showed declines to 19.6% of high school students and 4.7% of middle school students with current ENDS use; whether this is an artifact of the great societal disruptions from the global pandemic or represents a decreased trend remains to be seen.<sup>19</sup>

Because of the rapid evolution of ENDS, it is important to examine prevalence rates with other vaping products besides e-cigarettes such as e-hookahs (e-waterpipes). E-hookahs are a new category of vaping devices, introduced in 2014 and recently patented by Philip Morris, that are marketed as healthier alternatives to traditional hookah fruit-flavored tobacco smoking.<sup>20–22</sup> Findings from the nationally represented PATH study (Population Assessment of Tobacco and Health; 2014–2015) in children 12 to 17 years of age indicated that 7.7% were identified as ever-users of e-hookahs compared with 14.26% who were ever-users of ENDS products.<sup>23</sup>

### ENDS Product Use in Adults

In the United States, initial studies indicated a rapid increase in ever and current ENDS use among adults since 2010, with the vast majority of users being current or former cigarette smokers. Data from the NHIS (National Health Interview Survey) showed that although ever-use of ENDS among adults  $\geq 18$  years of age increased from 2014 to 2016 (12.6% to 15.3%), current use of ENDS decreased (3.7% to 3.2%).<sup>24</sup> Although the decrease in e-cigarette use was significant among current combustible cigarette smokers, the prevalence increased among former smokers and, it is important to note, never-smokers.<sup>24</sup> In 2016, data from the Behavioral Risk Factor Surveillance System showed that 1.2 million adults who never smoked combustible cigarettes were current ENDS users.<sup>25</sup> Dual use of ENDS and combustible cigarettes is common.<sup>26,27</sup> The 2016 Behavioral Risk Factor Surveillance System estimated that 54.6% of current ENDS users were current combustible cigarette smokers.<sup>25</sup> Recent analysis from the 2014 to 2018 NHIS data showed that young adults 18 to 24 years of age are using ENDS at high rates; current use increased from 5.1% to 7.6%, with large increases among never-smokers (1.5% to 4.6%) and former smokers (10.4% to 36.5%).<sup>28</sup>

### ACUTE HEALTH EFFECTS AND TOXICITY OF E-CIGARETTES AND VAPING PRODUCTS

ENDS aerosol condensate generated from different device types and products results in differential effects on measures of endothelial and pulmonary epithelial cell toxicity. The toxicological differences are likely due

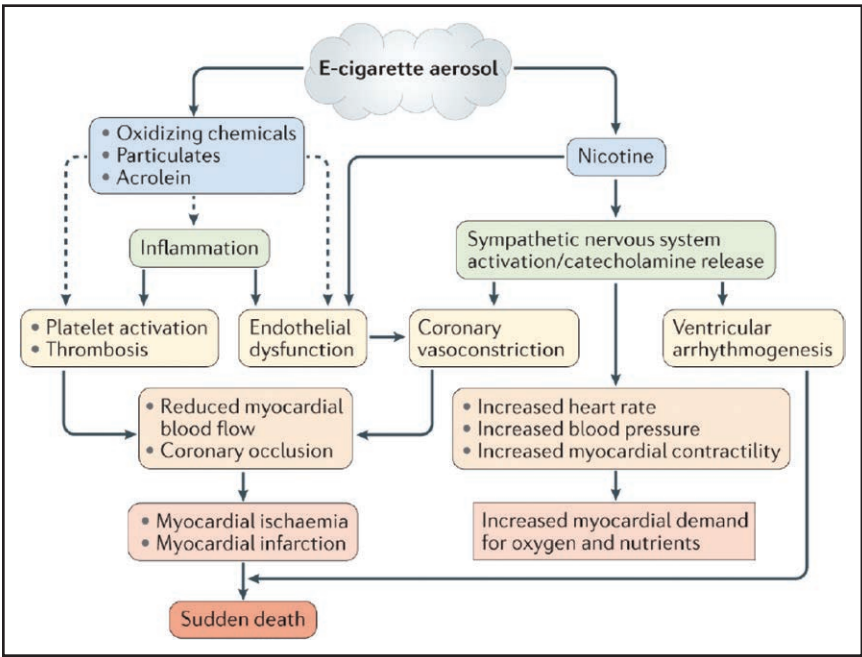
to differences in product characteristics such as nicotine levels and flavoring additives, which is supported by studies directly treating vascular and pulmonary epithelial cells with individual ingredients.<sup>29–31</sup> Nicotine is commonly found in ENDS products; however, not all ENDS products contain nicotine. Regardless of the presence of nicotine, many ENDS product constituents, including flavoring additives, hygroscopic carriers such as propylene glycol and glycerol, and metals (from the heating coil), have been shown to induce cardiopulmonary toxicity in animal and in vitro studies.

### Known Health Effects of Specific Components of Vaping Aerosols

#### Nicotine

Nicotine is an activator of the sympathetic nervous system, which has direct effects on the cardiovascular system.<sup>32</sup> In postganglionic sympathetic nerve terminals and the adrenal medulla, nicotine binds to  $\alpha 3\beta 4$  nicotinic acetylcholine receptors, triggering the release of catecholamines (norepinephrine, epinephrine).<sup>33</sup> Nicotine-stimulated catecholamine release by the sympathetic nervous system activates  $\beta$ -adrenergic receptors in the heart, resulting in increased heart rate, cardiac contractility, and workload (Figure 2).<sup>2,32</sup> Long-term overstimulation of the sympathetic nervous system results in cardiac remodeling, which promotes the development of heart failure and increases arrhythmogenesis.<sup>32</sup>

Nicotine also affects the vasculature by inducing vasoconstriction, resulting in elevated blood pressure and impaired wound healing in the microvasculature, which is mediated primarily through activation of the homomeric  $\alpha 7$  nicotinic acetylcholine receptor expressed on endothelial



**Figure 2. Overview of mechanisms by which electronic cigarette use might cause acute cardiovascular events.** Solid lines indicate known pathways; dashed lines indicate pathways of concern but for which there are no empirical data for confirmation. Reprinted with permission from Nature/Springer, Benowitz and Fraiman,<sup>2</sup> Copyright © 2017.

cells and  $\alpha 1$  receptors on vascular smooth muscle cells.<sup>32,33</sup> In vitro exposure of primary lung microvascular endothelial cells to nicotine increased primary lung microvascular endothelial cell permeability and impaired endothelial cell barrier function, which was likely mediated through oxidative stress–induced activation of p38 mitogen–activated protein kinase.<sup>34</sup> Nicotine has been shown to induce platelet-derived growth factor release, which induces cytoskeletal alteration in aortic smooth muscle cells.<sup>35</sup> In a randomized study of healthy younger smokers, acute use of a nicotine-containing ENDS had vascular hemodynamic effects suggestive of vascular remodeling and increased sympathetic activation of the cardiovascular system.<sup>36</sup> The findings of these studies may indicate cardiovascular changes consistent with the development of cardiovascular disease with nicotine inhalation from ENDS.<sup>36</sup>

### Propylene Glycol and Glycerol

Emerging evidence suggests that the nicotine solvents propylene glycol and glycerol have cardiopulmonary effects. High levels of propylene glycol (greater than the recommended maximum dose of 25 mg·kg<sup>-1</sup>·d<sup>-1</sup>) systemically can lead to metabolic acidosis, acute kidney injury, and a sepsis-like syndrome.<sup>37</sup> Glycol mixtures are used to create theatrical fog and smoke, and long-term occupational exposure is associated with higher reports of wheezing and chest tightness.<sup>38,39</sup> Short-term exposure to glycol mixtures is associated with acute dry cough and throat irritation, as well as decreased lung function in individuals with higher exposures.<sup>38,39</sup> The upper airway irritation associated with short-term occupational exposure to propylene glycol appears to be mediated by the sensory receptors TRPV1 and TRPA1, receptors also known to promote asthmatic inflammation and airway hyperreactivity in animal models.<sup>40–42</sup>

### Flavoring Additives and Sweeteners

Animal models and in vitro exposures of pulmonary and cardiovascular cells suggest that the flavoring additives have differing toxicities but are likely an important source of ENDS-induced cardiopulmonary toxicity. Sweeteners such as glucose and sucrose are commonly used in e-liquids and contribute to reactive aldehyde formation.<sup>43–45</sup> Reactive aldehydes are thought to be the primary contributors to combustible cigarette–induced cardiovascular disease and chronic obstructive pulmonary disease (COPD).<sup>46</sup>

Diacetyl and acetyl propionyl are flavoring additives that are associated with respiratory disease when inhaled during manufacturing.<sup>47</sup> Occupational exposure to diacetyl is associated with higher rates of chronic cough, shortness of breath, airway obstruction, and physician-diagnosed asthma and chronic bronchitis.<sup>48–50</sup> Cumulative diacetyl exposure positively correlates with spirometry abnormalities and decreased forced expiratory volume in the first second (FEV1) and has been shown

to result in the development of bronchiolitis obliterans, or popcorn lung.<sup>48–50</sup> Inhalation of diacetyl or 2,3-pentanedione resulted in fibroproliferation and partial occlusion of the airways, as well as peribronchial and perivascular infiltrations of immune cells in the lungs of exposed rats, consistent with airway remodeling.<sup>51</sup> Diacetyl and related flavoring additives continue to be found in e-liquids.

Other flavoring additives used in ENDS have also been shown to have pulmonary toxicity. Air-liquid interface exposure of bronchial epithelial cells to menthol-, coffee-, and strawberry-flavored e-liquids resulted in reduced cellular viability and increased inflammatory cytokine production.<sup>52</sup> Strawberry-flavored e-liquid elicited the greatest inflammatory response, similar to the effects observed with combustible cigarette smoke exposure.<sup>52</sup> Treatment of human bronchial epithelial cells for 24 hours with the flavoring additives acetoin (imparting a butter flavor), diacetyl, ortho-vanillin, and maltol (malt flavor) increased interleukin-8 secretion, suggesting that the flavor additives independently induce toxicity.<sup>53</sup> In vitro exposure of pulmonary epithelial cells and monocytes to aerosol generated with each of 8 flavored JUUL pods increased inflammatory cytokine levels, induced mitochondrial oxidant generation, increased DNA damage, and impaired pulmonary epithelial barrier function.<sup>54</sup> In addition, 2,5-dimethylpyrazine (imparts a chocolate flavor) treatment of human or mouse pulmonary epithelial cell lines decreased salt and water apical cellular transport, which may have implications for airway surface liquid homeostasis.<sup>55</sup>

A number of flavored e-liquids and commonly used flavoring additives have been shown to impair endothelial cell function in vitro.<sup>29,56</sup> Exposure of induced pluripotent stem cell–derived endothelial cells to cinnamaldehyde- and menthol-containing e-liquid decreased cell viability through increased apoptosis and decreased tube formation and wound healing (eg, scratch test) and upregulated the expression in inflammatory cytokines.<sup>29</sup> Endothelial cells exposed to cinnamaldehyde-containing e-liquid or RY4 e-liquid (sweet tobacco, caramel, and vanilla flavoring) for 48 hours had greater activation of the macrophage-like cells to a proinflammatory phenotype, suggesting that the flavored e-liquids promote endothelial cell inflammatory cytokine and chemokine secretion.<sup>29</sup>

Vanillin, menthol, cinnamaldehyde, eugenol (clove flavoring), and acetylpyridine (burnt flavoring) increased interleukin-6 expression and impaired A23187-induced nitric oxide production in human aortic endothelial cells.<sup>56</sup> Exposure to menthol or eugenol resulted in impaired nitric oxide production in response to A23187 and to a degree similar to the impairment observed in endothelial cells from combustible cigarette smokers.<sup>56</sup> In standing e-liquid, the interaction of flavor aldehydes with propylene glycol or glycerin may form toxic acetals.<sup>57,58</sup> These have the potential to have more respiratory irritation than the parent aldehyde molecules.<sup>57,58</sup>

## Metals

The heating elements of ENDS consist of metals that are released into the liquids and aerosols.<sup>59,60</sup> Rats exposed to aerosol generated with a nickel-chromium alloy atomizer that was free of nicotine, vitamin E acetate, or tetrahydrocannabinol developed labored mouth breathing, decreased activity, and wheezing.<sup>61</sup> Histological evaluation showed pneumonitis, pulmonary inflammation, and the accumulation of fibrin and inflammatory cells in the lumen of the trachea.<sup>61</sup> No such pulmonary alterations were noted for a similar device with a stainless steel atomizer, suggesting that the metal heating element may be a source of pulmonary toxicants.<sup>61</sup> A lifetime of chromium and nickel exposure as a result of daily inhalation of 2 mL e-liquid was used to estimate the risk of cancer and noncancer health effects, with chromium and nickel estimated to be the primary contributors.<sup>62</sup> Nickel is one of the few carcinogens found to be higher in ENDS than in combustible cigarettes.

## Acute Cardiovascular Effects of ENDS Product Use

Limited clinical studies have assessed the impact of ENDS on acute cardiovascular events. Large survey studies have focused on younger adults, who have a low incidence of heart attack and stroke. Prospective, longer-term studies of ENDS users of all ages with regard to cardiovascular events and symptoms such as palpitations and chest pain are needed.

Our current understanding of the deleterious cardiovascular effects is derived largely from short-term studies of ENDS users. Evidence indicates that there are acute changes in several hemodynamic measures, including increases in blood pressure and heart rate, with nicotine-containing ENDS use (Figure 2).<sup>63,64</sup> One systematic review estimates an increase of systolic and diastolic blood pressures of 2 mmHg and heart rate of 2 bpm for nicotine-containing ENDS products, but the relative impact compared with combustible cigarettes may be lower.<sup>65</sup> In fact, in a cross-sectional analysis of 39 747 participants in the Health eHeart Study, ENDS use compared with no ENDS and no cigarette use was associated with lower general health scores and greater proportions of those who responded “yes” to having chest pain, palpitations, coronary heart disease, and arrhythmia.<sup>66</sup> In addition, measures of heart rate variability show changes indicative of greater sympathetic tone after acute use, particularly in nicotine-containing ENDS versus non-nicotine-containing ENDS products.<sup>67,68</sup> Both acute nicotine-containing ENDS use and combustible cigarette use produce changes in ventricular repolarization, although the degree is more pronounced from the combustible product.<sup>69</sup>

ENDS products have been shown to impair endothelium-mediated dilation in the periphery, suggestive of reduced nitric oxide bioavailability.<sup>70–72</sup> Compared with sham vaping or vaping of vehicle alone (50:50 propyl-

ene glycol and glycerol), acute use of nicotine-containing liquid vaped through an e-cigarette device increased augmentation index, pulse-wave velocity, systolic blood pressure, diastolic blood pressure, and heart rate, suggesting vascular remodeling and sympathetic overactivation.<sup>70</sup> Although imaging studies have historically been limited in the evaluation of vascular dysfunction, a recent magnetic resonance imaging study showed reductions in blood velocity, vascular reactivity, and oxidation after ENDS use.<sup>72</sup> This study implies that the ENDS products may have a role in endothelial dysfunction and the development of cardiovascular disease.<sup>72</sup> E-hookah use has also been shown to have adverse effects on endothelial function, likely mediated by oxidative stress.<sup>73</sup> Higher arterial stiffness has been observed in ENDS users with an acute increase after use.<sup>64,74</sup> In a randomized study of healthy younger smokers, acute use of nicotine-containing ENDS products increased peripheral systolic blood pressure, heart rate, augmentation index, and pulse-wave velocity, but only when the e-liquid contained nicotine.<sup>36</sup> The effects of vaping a nicotine-containing e-cigarette or smoking a combustible cigarette resulted in similar elevations in pulse-wave velocity, suggesting increased arterial stiffness.<sup>36</sup>

In the coronary circulation, ENDS products lower the myocardial blood flow response to exercise without altering myocardial contraction or relaxation.<sup>63,75</sup> Long-term users of ENDS products had higher arterial stiffness but similar endothelial function compared with nonsmokers, suggesting that the endothelial effects may be largely present in the postuse state.<sup>76</sup> Studies have shown lower blood pressure, lower heart rate, and markers of improved vascular function in combustible cigarette users who switch to ENDS products.<sup>77–79</sup> Cigarette smoke has been shown to increase platelet activation, reactivity, adhesion aggregation, and inflammation, ultimately driving ischemic events such as myocardial infarction.<sup>80</sup> Platelet activation, reactivity, and aggregation have been shown to be increased after exposure to e-liquid aerosols in both nicotine-containing<sup>81–84</sup> and non-nicotine-containing substances.<sup>85,86</sup>

Studies comparing ENDS products that contain nicotine with nicotine-free products suggest that many of the acute vascular effects relate to nicotine exposure.<sup>67,74</sup> Nicotine-free aerosol has been shown to affect endothelial barrier function, postulated to be from the e-cigarette aerosol constituent acrolein.<sup>34,86</sup> The studies have included predominantly ENDS users who have a history of combustible cigarette use. A study of young healthy adults who never used combustible cigarettes did not observe any acute change in endothelial function.<sup>87</sup>

Studies of circulating biomarkers indicative of cardiovascular risk have also shown changes with acute ENDS use. Measures of oxidative stress increase after ENDS use, although less than with combustible cigarette use.<sup>71</sup> There appeared to be residual impact on inflammation and

oxidative stress in e-cigarette users even after a period of nonuse.<sup>88</sup> In a population-based study, inflammatory and oxidative stress biomarkers were elevated in both combustible cigarette users and dual combustible cigarette and ENDS users but not in exclusively ENDS users.<sup>89</sup>

There have been minimal studies on the acute impact of ENDS products in individuals with established cardiovascular disease who may be more at risk to negative health effects of ENDS. Whether the acute known cardiovascular effects of ENDS, particularly those containing nicotine, translate into heightened cardiovascular risk needs to be assessed fully. The presence of any acute cardiovascular effects of ENDS suggests that ENDS use is not benign and raises the possibility that adverse impact could accumulate over time.

### General Cerebrovascular Effects of Smoking and E-Cigarettes

Combustible cigarette smoking increases risk of neurological disorders, including stroke<sup>90</sup> and vascular dementia<sup>91</sup> among others. Studies show that tobacco smoking can affect vascular endothelial function and the blood-brain barrier<sup>92</sup> with long-term consequences to the brain. More intrinsic mechanisms include increased oxidative stress and inflammation,<sup>93</sup> promotion of type 2 diabetes,<sup>94</sup> synaptic and electric activity alterations, and ultimately behavioral changes.<sup>95–97</sup> Tobacco smoking increases the risk of stroke, and nicotine exacerbates poststroke brain damage.<sup>98</sup> Studies in animal models have shown that nicotine can cause pathological effects similar to those observed with tobacco smoke such as oxidative damage and cerebrovascular impairments that lead to stroke and exacerbate ischemic injury.<sup>99–101</sup>

Similar to tobacco smoke, ENDS products increase vascular stiffness, increase endothelial microparticles, and promote reactive oxygen species through the NADPH oxidase pathway.<sup>95</sup> Another effect of nicotine and ENDS exposure is a decrease in brain glucose utilization.<sup>102</sup> Prolonged exposure to nicotine leads to an increase in GLUT1 expression in the brain, with a consequent decrease in glycolysis.<sup>102</sup> This glucose deprivation state may enhance ischemic brain injury or stroke risk.<sup>102</sup> Nicotine may lead to brain mitochondrial dysfunction because  $\alpha 7$ -nAChR is highly permeable to  $\text{Ca}^{2+}$ , a key player in both synaptic plasticity and ischemia-induced cell death.<sup>103–105</sup>

The systemic and metabolic effects of ENDS products are accompanied by changes in synaptic and electric activity in the brain. Beyond the addictive effects of nicotine (see later discussion), changes in glutamatergic receptors were observed after months of ENDS exposure: In the nucleus accumbens (shell), mGluR1 was upregulated, whereas mGluR5 and the glutamate transporter-1 were downregulated,<sup>106</sup> signaling potential addictive effects. The effect of ENDS on the hippocampus was an upregulation of mGluR5 and the glutamate transporter-1, which could elicit potential pathologi-

cal consequences for cognitive performance or worsen the pathology of cerebral ischemia.<sup>106</sup> Furthermore, because the hippocampus is susceptible to ischemia, these effects could worsen the pathology of cerebral ischemia. The effects of nicotine on stroke patients are more prominent in women than men, thought to result partially from both nicotine and oral contraceptives being risk factors for peripheral thrombus formation.<sup>107–110</sup> The mechanisms for smoking and nicotine with oral contraceptive-induced thrombogenesis remain poorly understood. Nicotine exposure from ENDS products can inhibit aromatase enzyme activity, which catalyzes the conversion of androgens into estrogens.<sup>111</sup> In female rats, it was confirmed that long-term nicotine exposure reduced endogenous  $17\beta$ -estradiol levels.<sup>112</sup> Endogenous and exogenous  $17\beta$ -estradiol is known to protect the brain from ischemic injury in female rats.<sup>113–116</sup> Thus, worsening of cerebral ischemic outcomes by nicotine in women may involve decreases in estrogen levels.

### Acute Pulmonary Effects of ENDS Product Use

The understanding of the acute pulmonary effects of ENDS is highlighted by several epidemiological studies on respiratory symptomatology, physiological impact on pulmonary function, and animal or molecular studies. Large surveys of teens, adolescents, and adults have reported increased odds of chronic cough, phlegm, bronchitis, and dyspnea, especially with simultaneous cigarette use.<sup>66,117–121</sup> Large survey data in North America, Asia, and Europe demonstrated higher self-reported history of asthma, asthma exacerbations, and COPD.<sup>122,123</sup> Upper respiratory effects such as nose, mouth, throat, and airway irritation have been reported with ENDS use.<sup>124,125</sup> Prospective cohort studies are ultimately needed to assess the longer-term adverse consequences of ENDS products.

The acute effects of ENDS products on airway obstruction and lung volumes have been examined with pulmonary function testing. Some studies show evidence of airflow obstruction after an acute vaping session, with decreased FEV1, forced expiratory ratio (FEV1/forced vital capacity [FVC]), and forced expiratory flow 25% to 75% (FEV25%–75%).<sup>126,127</sup> Other studies did not demonstrate an acute change in airway obstruction, lung volumes, or diffusion capacity after ENDS use compared with control subjects.<sup>128–131</sup> Patients with preexisting airway disease may be at higher risk from acute worsening of airway obstruction with ENDS use.<sup>132</sup> Similar to cigarette smoking, some studies show a decreased fraction of exhaled nitric oxide with ENDS use.<sup>133,134</sup> Other measures of airway resistance such as impedance, respiratory resistance, and peripheral airway resistance were increased after ENDS product use for 5 minutes.<sup>132,133</sup>

In terms of evidence of direct lung injury from ENDS products, sporadic cases were reported before the outbreak of e-cigarette or vaping product use-associated lung injury (EVALI) in 2019 (see the section on EVALI).

Cases of eosinophilic pneumonitis,<sup>135,136</sup> hypersensitivity pneumonitis,<sup>137</sup> organizing pneumonia,<sup>138</sup> diffuse alveolar hemorrhage,<sup>139</sup> respiratory bronchiolitis–associated interstitial lung disease,<sup>140</sup> and acute lung injury were reported after ENDS use.<sup>141</sup> Many flavoring agents in e-liquids have known toxic effects on the lungs, particularly diacetyl (bronchiolitis obliterans, or popcorn lung),<sup>142</sup> 2,3 pentanedione,<sup>143,144</sup> vanillin, and cinnamaldehyde.<sup>145</sup> Whether these early cases of severe respiratory disease were individual susceptibility to ENDS use or reflected a particular agent remains unclear.

Several studies examined markers of airway injury and inflammation in otherwise healthy e-cigarette users. Healthy ENDS users were found to have erythematous and irritable airway mucosa.<sup>146</sup> Increased airway and bronchial epithelial MUC5AC (mucin-5AC) was found in ENDS users.<sup>146–148</sup> Increased mucin is a biomarker of chronic bronchitis in patients with declining lung function in COPD.<sup>31,149</sup> Sputum from ENDS users has shown increased neutrophil activation, myeloperoxidase, neutrophil elastase, and proteinase-3.<sup>148,150</sup> Increased proteases in the lung are thought to be a significant driver of lung damage in COPD from tobacco smoking and play a role in tumor progression.<sup>151–153</sup> Neutrophils collected from long-term ENDS users have greater neutrophil extracellular trap formation, a process that can injure the lung, compared with nonsmokers and cigarette smokers.<sup>148,154</sup> Studies with volunteers exposed to short sessions of vaping revealed both increased blood endothelial microparticles (marker of alveolar capillary injury) and lung-specific protein CC16 (secreted from club cells near terminal bronchioles suggestive of small airway injury), suggesting molecular changes.<sup>129,155</sup> Common molecular markers of lung injury and airway inflammation have been observed with exposure to ENDS.

ENDS use has been linked to an increased susceptibility to respiratory infection. A study with 30 healthy volunteers exposed to e-cigarette aerosol demonstrating decreased cough sensation, in combination with basic and animal studies showing ciliary dysfunction, suggested susceptibility to respiratory infections.<sup>156,157</sup> Nasal scrapings from vapers, smokers, and nonsmokers showed suppression of immune and inflammatory response genes in smokers and vapers.<sup>158</sup> In bronchoalveolar lavage of healthy nonsmokers, after brief ENDS aerosol exposure, altered gene expression was seen in alveolar macrophages, notably in inflammation-related genes.<sup>129</sup> With the COVID-19 pandemic, several epidemiological studies have assessed the effect of ENDS use on the development of or severity of COVID-19, but results have been conflicting.<sup>159</sup>

In most studies of the acute physiological effects of vaping, clinical changes are mild, similar to findings demonstrated in the early phase of cigarette adoption in younger healthy cigarette smokers in 1920s. The most detrimental long-term effects of cigarettes, borne out after years of smoking, were not identified more conclusively until decades later. Modern assessment at the

molecular and cellular scale was not available in the early 20th century. Increasing evidence suggests that ENDS products have significant acute injurious effects on the airway epithelium, increased pulmonary inflammation, and respiratory immune suppression (Figure 3).<sup>8</sup>

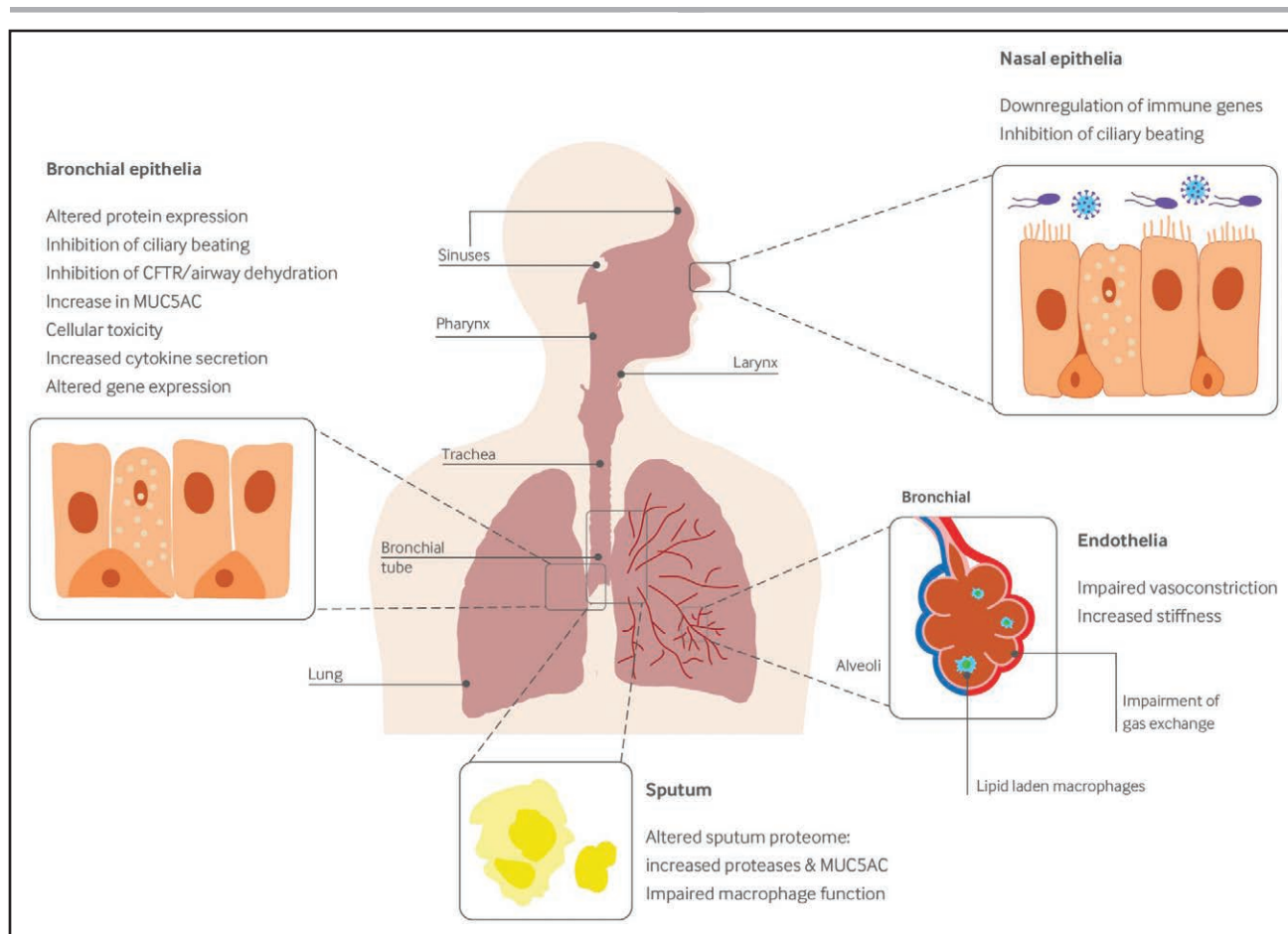
### **Acute Effects of ENDS Product Use During Pregnancy**

The use of ENDS products during pregnancy is prevalent among expectant mothers who smoke throughout pregnancy and who are using ENDS products to quit combustible cigarette smoking. For women who are trying to reduce the stigma or health concerns during pregnancy, ENDS use is often perceived as a safer alternative compared with combustible cigarette smoking.<sup>160</sup> Focus groups of pregnant smokers and smokers planning for pregnancy revealed that ENDS products were thought to have several advantages over cigarettes, including lower cost, the perception of being smoke-free, availability in appealing flavors, and fewer side effects despite the overall opinion that the use of any tobacco product was harmful during pregnancy.<sup>160</sup> It is important to note that many pregnant mothers become dual users, continuing to use both tobacco cigarettes and ENDS products throughout pregnancy.<sup>161</sup> There is a gap in knowledge in that studies on gestational health effects of ENDS products on the mother and on fetal development are lacking. Similarly, data on long-term health effects of ENDS exposure on the fetus after birth or on the mother are lacking.

Data in animal models reveal that ENDS products can be toxic to both the mother and the fetus. Reduced placental and cord blood flow has been reported with maternal ENDS use, along with growth restriction in the fetus and growth delay in the postnatal period.<sup>162</sup> In animal models, nicotine exposure during pregnancy leads to fetal growth deficits, systemic inflammation, fetal pulmonary and cardiac dysfunction, and cognitive issues in the newborn.<sup>163,164</sup> Perinatal exposure to nicotine results in angiotensin II–mediated vascular changes in offspring.<sup>165</sup> Furthermore, exposure in pregnancy to propylene glycol/vegetable glycerin, with or without nicotine, resulted in cognitive deficits in mice pups.<sup>164</sup> Lung development in newborns can be severely impaired from in utero exposure to nicotine.<sup>163</sup> The fetal cardiopulmonary health effects of in utero ENDS exposure, especially those containing nicotine, may be similar to those of combustible cigarettes and warrants further investigation.

### **Acute Addictive Effects of Inhaled Nicotine Products**

Most ENDS products contain the constituent nicotine, which is the main mechanism for dependence. After brief, short-term inhalation of nicotine with an ENDS, brain nicotine concentrations have been shown to increase in adult humans.<sup>166</sup> Positron emission tomography imaging shows that e-cigarettes deliver both relatively high (ie, 36 mg/mL) and low (8 mg/mL) nicotine level occupancy of



**Figure 3. Reported effects of vaping on the human pulmonary system.**

CFTR indicates cystic fibrosis transmembrane conductance regulator protein; and MUC5AC, mucin 5, subtypes A and C, tracheobronchial/gastric protein. Reprinted from Gotts et al.<sup>8</sup> Copyright © 2019 The Authors. Published on behalf of the authors by BMJ Publishing Group Ltd. This is an Open Access article distributed in accordance with the terms of the Creative Commons Attribution (CC BY 4.0) license, which permits others to distribute, remix, adapt and build upon this work, for commercial use, provided the original work is properly cited. See <http://creativecommons.org/licenses/by/4.0/>.

the  $\beta^2$  nicotinic acetylcholine receptors in the brain, levels of occupancy similar to those achieved with smoking a combustible cigarette.<sup>167</sup>

The acute addictive effects of nicotine-containing ENDS products stem from the immediate reinforcing effects produced and from the relief from symptoms associated with nicotine abstinence in those with existing nicotine dependence.<sup>168</sup> In terms of reinforcing effects, compared with non-nicotine-containing ENDS products, ENDS products containing nicotine demonstrated increased acute satisfaction and pleasant sensations, as well as feelings of alertness.<sup>169</sup> In another study assessing the dependence-reinforcing effects of ENDS, after e-cigarette use in tobacco product users, feelings of pleasantness and satisfaction were increased.<sup>170</sup> Acute exposure to ENDS products in experimental studies decreases feelings of nicotine craving among nicotine users,<sup>171</sup> and compared with ENDS products with no nicotine, nicotine-containing ENDS products had superior reduction of craving.<sup>172</sup> Newer generations of ENDS products appear to have higher effects on craving reduction and positive

associated subjective effects than older generations.<sup>171</sup> The product characteristics of newer ENDS products such as device power and voltage appear to increase nicotine delivery, affecting the effectiveness of nicotine delivery of these devices and thus the acute addictive effects.<sup>173</sup>

## E-CIGARETTE OR VAPING PRODUCT USE-ASSOCIATED LUNG INJURY

Vaping can cause a wide array of severe adverse health effects that include nicotine poisoning, trauma from device battery explosions, and injury to the gastrointestinal, cardiovascular, and neurological systems.<sup>174,175</sup> Severe respiratory effects of vaping in case reports include status asthmaticus<sup>176</sup> and pneumothoraces.<sup>177</sup> Diffuse parenchymal lung disease had been reported for several years in case reports.<sup>135–138,140,141</sup> With the rapid adoption of vaping among the public, particularly youth, a new clinical entity, EVALI, was first recognized in August 2019.<sup>178</sup> When the US Centers for Disease Control and Prevention stopped reporting

EVALI cases on February 18, 2020, there were 2807 hospitalizations and 68 deaths resulting from EVALI in all 50 states within the United States.<sup>179</sup> Clinically, patients develop hypoxemia requiring supplemental oxygen and signs of systemic inflammation indicated by leukocytosis.<sup>180,181</sup>

## Clinical Presentation

Three types of symptoms commonly present across EVALI cases: prominent gastrointestinal, general systemic (fever and fatigue), and respiratory symptoms.<sup>182–187</sup> Case series demonstrate that nearly all patients have presented with respiratory complaints.<sup>182–187</sup> Most patients required some form of respiratory support, ranging from supplemental oxygen, noninvasive ventilation, and mechanical ventilation to extracorporeal oxygenation.<sup>182–187</sup> Gastrointestinal symptoms also have been a common chief complaint, affecting >80% of patients.<sup>183</sup> Constitutional symptoms, including subjective fever, chills, and fatigue, have been common.<sup>182–184</sup> Definitions of confirmed and probable EVALI are shown in [Supplemental Table 1](#).<sup>188</sup> The long-term effects of EVALI have been minimally studied, but 1 cohort of 41 patients with 1-year outcomes reported 24% of patients requiring inpatient readmission, 4.9% 1-year mortality, and 45% with continued symptoms on outpatient follow-up. Although most imaging findings were improved or resolved on follow-up, 75% of patients with pulmonary function testing had abnormalities after EVALI.<sup>189</sup>

## Management

Management of EVALI is anecdotal and based on limited experience from treating similar disease processes. No clinical trials have tested therapies. In general, treatment is supportive. According to several reported case series, most patients described as having EVALI were initially managed in inpatient services.<sup>182–185</sup> In some series, more than half have presented with severe illness requiring intensive care unit admission.<sup>182–185</sup> Most patients (78%–100%) have required some degree of supplemental oxygen during hospital admission.<sup>183–186</sup> Severe respiratory failure has been common, with a maximal level of respiratory support that has included high-flow nasal cannula support (up to 47% in 1 study), noninvasive positive-pressure ventilation (up to 30%), and endotracheal intubation and invasive mechanical ventilation (up to 22%).<sup>182–184</sup> Severe cases of EVALI have required veno-venous extracorporeal membrane oxygenation, with 1 case proceeding to lung transplantation.<sup>176,191,192</sup> According to another review of 169 published articles on EVALI, most patients (95%) required hospitalization, and the majority (84%) received glucocorticoids.<sup>193</sup>

Glucocorticoids were used in many case series and may be beneficial in the treatment of EVALI.<sup>182,187,194,195</sup> Dosing regimens varied widely.<sup>185</sup> Glucocorticoid admin-

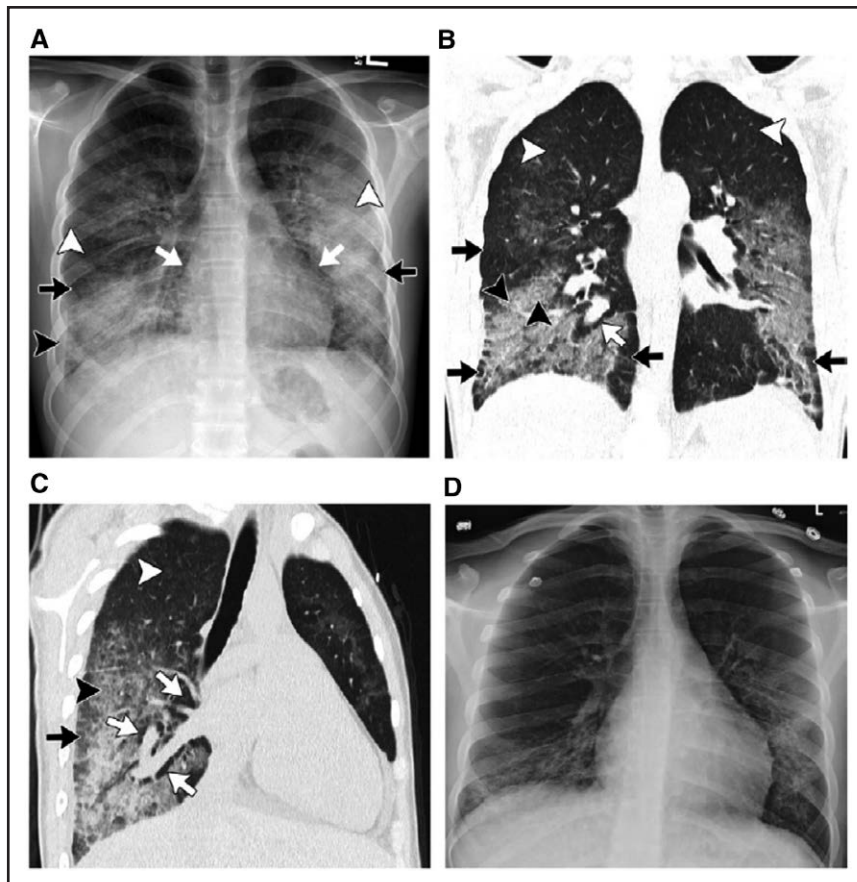
istration may be warranted in severely ill patients when it is considered safe and feasible and concomitant infection has been ruled out; however, no definitive clinical trials have assessed their effectiveness. Patients have recovered without receiving glucocorticoids.<sup>183–185</sup> Most patients had a full microbiological and viral infection workup.<sup>182–187</sup> Between 78% and 100% of patients received empirical antibiotics therapy, at least initially, in case series.<sup>182–187</sup> Cases of recurrent EVALI with persistent ENDS use have been reported; therefore, cessation is paramount for these patients.<sup>189</sup>

## Radiographic and Pathological Findings of EVALI

Because EVALI is often a diagnosis of exclusion, abnormal pulmonary imaging findings are typically required.<sup>190</sup> Pathological features associated with this pulmonary injury are consistent with a chemical-induced pneumonitis.<sup>196</sup> The pattern of acute lung injury is typical of organizing pneumonia, diffuse alveolar damage, or both. Acute eosinophilic pneumonia and diffuse alveolar hemorrhage are less common findings.<sup>190</sup> Typical computed tomography imaging features of EVALI-related organizing pneumonia consist of diffuse, symmetric, bilateral ground-glass opacities with subpleural sparing (Figure 4).<sup>190</sup> Thickening of the interlobular septum and centrilobular nodules is also frequently present.<sup>190</sup> Patients who develop diffuse alveolar damage demonstrate greater severity of pulmonary findings, consistent with more advanced disease. Imaging findings commonly include volume loss and consolidation or ground-glass opacities with a lower lobe predominance. As disease progresses, volume loss and architectural distortion increase.<sup>190</sup> Progression to chronic pulmonary disease is characterized by severe scarring. This progression can result in severe respiratory failure and even death.<sup>176,179,191,192</sup> Patients who survive can have chronic pulmonary dysfunction, and in rare cases, lung transplantation has been performed for EVALI.<sup>176,182,191,192</sup> Computed tomography imaging can be helpful in patients with chronic pulmonary injury to assess the degree of scarring and to diagnose complications associated with respiratory failure. These complications can include superimposed infection, formation of bullae, and development of pneumothorax.<sup>190</sup>

## Proposed Mechanism of Action for EVALI

Vitamin E acetate has been implicated as the most likely causal factor for EVALI in the patients investigated in the 2019 US epidemic.<sup>181</sup> Vitamin E acetate is used as a thickening agent in some nicotine-containing e-cigarettes but mostly those containing tetrahydrocannabinol oil. Vitamin E acetate was found in the bronchial alveolar lavage of 49 of 51 patients (94%) with EVALI.<sup>181</sup> No other toxicant in the screening panel was identified, leading



**Figure 4. Images show electronic cigarette or vaping product use–associated lung injury with organizing pneumonia pattern secondary to vaping tetrahydrocannabinol in an 18-year-old man.**

**A**, Posteroanterior radiograph shows midlung and lower lung consolidation and opacity bilaterally. Small right pleural effusion (black arrowhead) and septal thickening (white arrowhead) are seen. There is conspicuous sparing of cardiac borders (white arrows) as well as subpleural portions of lung (black arrows). **B**, Coronal and **C**, sagittal oblique images from CT nicely illustrate radiographic findings with mild and lower lung–predominant ground-glass opacity with few areas of consolidation. Prominent subpleural and perilobular sparing is present (black arrows). In addition, there is conspicuous sparing of peribronchovascular interstitium, best illustrated around larger pulmonary arteries and veins (white arrows). Hazy upper lobe–predominant ground-glass centrilobular nodules are present bilaterally (white arrowheads). In addition to thickening of interlobular septa, there are few areas with thickening of intralobular septa creating “crazy paving” pattern (black arrowheads). **D**, Three days after initiation of steroids, patient showed dramatic clinical and radiographic improvement. Reprinted with permission from Kligerman et al.<sup>190</sup> Copyright ©2020 RSNA.

to the suggestion that vitamin E acetate may be linked to the development of the 2019 EVALI epidemic in the United States.<sup>181</sup> Both untargeted and targeted mass spectrometry evaluation of 38 e-liquid cartridges from patients who had developed EVALI revealed that 64% of the e-liquids contained vitamin E acetate.<sup>198</sup> The absence of vitamin E acetate in several cases and products suggests that additional product characteristics could contribute to the development of EVALI.

Elevated albumin, CD45<sup>+</sup> cells, and lipid-laden macrophages were found in the bronchial alveolar lavage fluid of mice exposed to 2 weeks of propylene glycol and glycerol with vitamin E acetate compared with filtered air–exposed animals.<sup>199</sup> Similar alterations were observed in animals exposed to only propylene glycol and glycerol, but not to the same extent as the animals that were exposed to vitamin E acetate.<sup>199</sup> Overall, evaluation of vitamin E acetate exposure in animal models shows a pulmonary pathology similar to that observed in patients

with EVALI, suggesting that vitamin E acetate may be one of the contributors to the disease development.

### The Future of EVALI

Emergency department visits related to EVALI declined sharply by early 2020, thought to be through a combination of increased public awareness about informal vaping products through rapid public health response (before the COVID-19 pandemic), removal of vitamin E acetate from some products, and law enforcement actions related to illicit products. The emergence of the COVID-19 pandemic and the overlap of many symptoms of EVALI and COVID-19 also likely contributed to less recognition of EVALI. The National Heart, Lung, and Blood Institute of the National Institutes of Health convened a working group on October 23, 2019, to identify the most relevant and urgent research priorities and challenges in EVALI.<sup>200</sup> One key goal was to lay out

**Table 1. National Institutes of Health Workshop on EVALI Conclusions From October 2019<sup>200</sup>**

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Key points of agreement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| E-cigarette or vaping devices have been used to generate aerosols containing numerous active substances for inhalation, including nicotine, THC, and CBD.                                                                                                                                                                                                                                                                                                                                                 |
| The etiology of EVALI remains unknown. Available data from the CDC point to an association with oil-based ingredients, such as vitamin E acetate, added to THC liquids before vaping or dabbing (inhaling potent vapors from concentrated marijuana oil), but a specific cause for EVALI has not been proven mechanistically.                                                                                                                                                                             |
| Vaping devices have not been shown to be safe for long-term use. The short- and long-term toxicities of inhaling aerosols generated from liquids containing vegetable glycerin, propylene glycol, nicotine, or flavors are unknown. Inhaling aerosols generated from THC- or CBD-containing liquids, which often contain additional chemical components, also have unknown health effects. Thus, elucidating their long-term respiratory, cardiac, and cancer health effects is a public health priority. |
| Physicians and other health care professionals need further guidance on how to take an accurate and detailed inhalant history. Acquisition of such data from across the country will help track EVALI cases and other vaping-related health effects.                                                                                                                                                                                                                                                      |
| Key priorities for future research                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| National and international registries of all patients with EVALI should be created that could be accessed by researchers and clinicians to track those affected and assess outcomes accurately.                                                                                                                                                                                                                                                                                                           |
| We suggest that 3 cohorts of subjects be identified and followed up over time: (1) healthy vapers/dabbers, (2) patients with severe EVALI, and (3) patients with nonsevere (mild, moderate, subacute, and chronic) EVALI. Groups would be powered to identify causal agents, short- and long-term effects on lung function, overall health, and biomarkers of EVALI in both target organs (eg, the lung) and accessible biofluids (eg, blood, urine).                                                     |
| Acquisition and analysis of biospecimens from humans affected by EVALI (whole blood, tracheal aspirates or BAL, lung biopsy specimens, urine, autopsy specimens) will help profile inflammatory and fibrotic changes, determine whether a heretofore unknown viral or bacterial pathogen may be a causal agent, and gain an overall understanding of EVALI pathogenesis.                                                                                                                                  |
| Research on vaping effects to date may not have identified EVALI because of multiple limitations, including but not limited to lack of access to and hence testing of THC liquids used in vaping and dabbing, rapid evolution of products on the market, inadequate dosing or exposure to e-cigarette aerosol in animal models, and an absence of studies in which multiple device types and liquids are applied to models to mimic use patterns in the affected demographic.                             |
| The occurrence of EVALI highlights the importance of broadening e-cigarette toxicity questions beyond comparators to smoking and smoking-related diseases because vaping has disease risks different from smoking.                                                                                                                                                                                                                                                                                        |
| Systematic and comprehensive laboratory-based studies should be conducted examining the toxicology and health effects of THC-containing vaping products to provide insights into potential causes of EVALI.                                                                                                                                                                                                                                                                                               |

BAL indicates bronchoalveolar lavage; CBD, cannabidiol; CDC, Centers for Disease Control and Prevention; e-cigarette, electronic cigarette; EVALI, electronic cigarette or vaping product use–associated lung injury; and THC, tetrahydrocannabinol.

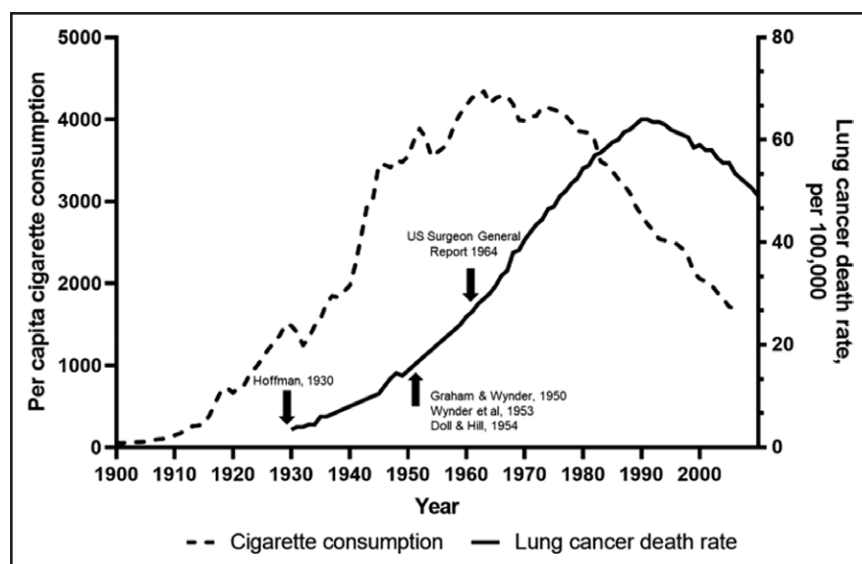
Adapted with permission from Crotty et al.<sup>200</sup> Copyright © 2020 American Thoracic Society.

future research priorities and to recommend strategies for the diagnosis, treatment, and prevention of the disease (Table 1).<sup>200</sup> There was consensus that long-term outcomes in EVALI are unknown<sup>200</sup>; in the ensuing years, reports emerged on long-term outcomes of EVALI,<sup>189</sup> but more data are needed.

## CHRONIC HEALTH EFFECTS OF E-CIGARETTES AND VAPING PRODUCTS

E-cigarettes and vaping were introduced to the US market 15 years ago, with widespread adoption only in the past decade.<sup>11,13</sup> We do not yet know the long-term health effects of these products. Tobacco use is a major preventable cause of death; however, this was not appreciated until many years after cigarette smoking became widespread. Before the adoption of tobacco products, lung cancer was rare.<sup>201</sup> Increasing incidence of lung cancer in actuarial studies was not noted until 1930 (Figure 5).<sup>201–206</sup> Definitive scientific evidence associating cigarette smoking and lung cancer was not reported until the 1950s.<sup>203–205</sup> In 1964, the US Surgeon General report on tobacco and health attributed the increase in lung cancer to cigarette smoking.<sup>206</sup> Only then did cigarette smoking per capita begin to decline. With the delayed development of chronic disease from smoking, lung cancer deaths did not begin to fall accordingly until decades after the 1964 report (Figure 5).<sup>201–206</sup> Similarly, declines in mortality from myocardial infarction and stroke in recent decades have been attributed in part to the significant decline in cigarette smoking in the past 50 years.<sup>207</sup> Supposing similar time delays for the presentation of chronic disease from cigarettes (eg, lung cancer, COPD, or vascular disease) and for ENDS, epidemiological increases in disease prevalence would not be expected to be observed for years after ENDS products were introduced to the US market. It is critical to assess the limited amount of long-term clinical and animal data that emerge to better understand the long-term effects of ENDS use. The National Academies of Sciences, Engineering, and Medicine reviewed the health literature of ENDS products in 2018 and provided an evidence-based summary of health concerns.<sup>208</sup> The report found that there was no available evidence on whether ENDS use is associated with clinical cardiovascular outcomes such as coronary heart disease, stroke, and peripheral artery disease.<sup>208</sup> There was insufficient evidence that ENDS use was associated with long-term effects on heart rate, blood pressure, and cardiac function.<sup>208</sup> The report concluded that there was no available evidence on whether ENDS use causes respiratory diseases in humans, moderate evidence that ENDS use is associated with an increase in asthma exacerbations, and limited evidence of adverse effects of ENDS exposure on the respiratory system from animal and in vitro studies.

Few studies have evaluated the chronic cardiovascular effects of ENDS because the products have been available only for the past 10 to 15 years. In a murine model, 8 months of ENDS exposure increased arterial stiffness, reduced stimulated vascular relaxation, and enhanced sensitivity to vasoconstriction, similar to vascular alterations induced by cigarette smoking.<sup>209</sup> No effects were observed on more acute physiological markers such as heart rate, stroke volume, or cardiac output with e-cigarette



**Figure 5. Tobacco use vs lung cancer death rates in the United States.**

Adapted from American Cancer Society. Cancer Facts & Figures 2013. Atlanta, GA: American Cancer Society; 2013.<sup>202</sup>

exposure.<sup>209</sup> Mice exposed to ENDS aerosol for 3 to 6 months have increased inflammation, organ fibrosis, and elevated systolic blood pressure compared with mice exposed to air<sup>210</sup>; another study showed that 3 months of ENDS exposure showed resulted in mutagenic DNA adducts in tissues, including the heart.<sup>211</sup> In the aforementioned studies, observed chronic toxicity of ENDS could be attributed, at least in part, to nicotine.<sup>210,211</sup>

Limited clinical data are available for chronic ENDS effects on the cardiovascular system. One cross-sectional case-control study showed that compared with nonvaping control subjects, regular ENDS users had a shift in autonomic balance toward sympathetic predominance, higher low-density lipoprotein oxidation, and elevated markers of inflammation, suggestive of an association with increased cardiovascular risk.<sup>68</sup> Activation of the splenic axis after long-term exposure to cigarettes and ENDS products has been observed in healthy young adults.<sup>212</sup> Activation of the splenic axis can be proinflammatory and is associated with the development of atherosclerosis and acute myocardial infarction.<sup>212</sup> The NHIS data of 2014 and 2016 were used to examine the cross-sectional association of ENDS use, cigarette smoking, and myocardial infarction.<sup>213</sup> With the use of a single logistic regression model that included demographics and medical history, daily ENDS use was independently associated with increased odds of having had a myocardial infarction (odds ratio, 1.79 [95% CI, 1.20–2.66]).<sup>213</sup> Although current evidence suggests possible cardiovascular risk from long-term exposure to ENDS, more long-term animal exposure model studies, population studies, and prospective clinical studies are needed.

A recent longitudinal analysis of the adult PATH study waves found a statistically significant association between former or current ENDS use at baseline and the development of incident respiratory disease (COPD, chronic bronchitis, emphysema, or asthma) at a follow-up

of  $\approx 2$  years (adjusted OR, 1.31 and 1.29, respectively).<sup>214</sup> The majority of the investigations to study long-term respiratory effects associated with ENDS use were conducted in either current or previous cigarette smokers. A 1-year study found that FEV25%–75% improved in smokers who quit combustible cigarettes but other spirometry parameters (FEV1, FVC, or FEV1/FVC) were unchanged, regardless of complete cessation of ENDS use, suggesting that e-cigarette use in smokers may result in some improvement in small airway obstruction.<sup>215</sup> In observational cohorts of patients with COPD who were current or former smokers 45 to 80 years of age (COPD Gene [Chronic Obstructive Pulmonary Disease Genetic Epidemiology],  $n=3536$ ; SPIROMICS [Subpopulations and Intermediate Outcome Measures in COPD Study],  $n=1060$ ), ENDS use was associated with a greater decline in lung function, higher nicotine exposure, and greater risk of COPD exacerbation.<sup>216</sup> No evidence of harm reduction was observed with ENDS use compared with combustible cigarettes.<sup>216</sup> Conversely, another study found improvements in COPD exacerbation and test scores during a 6-minute walk test with ENDS use compared with combustible cigarette use over 3 years.<sup>217</sup> This study was small ( $n=22$  per group) and lacked a comparison with never-smoking patients, limiting the conclusions that can be drawn on the safety of ENDS use.<sup>217</sup>

A limited number of studies are available evaluating e-cigarette use and exposure among patients with chronic pulmonary conditions. A randomized controlled trial examined the effects of passive ENDS aerosol exposure in patients who had COPD who were nonsmokers.<sup>218</sup> Passive e-cigarette exposure resulted in alterations in exhaled surfactant protein A and increases in certain plasma proteins that could indicate inflammation.<sup>218</sup> In a large cross-sectional telephone survey of never-smokers, the odds of self-reported asthma were higher in current ENDS users compared with nonusers.<sup>219</sup> Similar survey

studies have shown higher self-reported asthma and COPD in ENDS users.<sup>124,125</sup> Acute ENDS exposure (5 minutes) does appear to worsen respiratory mechanics immediately after use in patients with asthma.<sup>132</sup> House dust or ovalbumin challenge in mice exposed to ENDS aerosol resulted in various alterations in airway hyperresponsiveness and airway inflammation that were at least partially dependent on flavoring and nicotine.<sup>220–222</sup>

No studies on ENDS product use have been performed in cohorts of patients with cystic fibrosis, interstitial lung disease, or pulmonary hypertension. A recent report found that vaporized ENDS liquids can induce CFTR (cystic fibrosis transmembrane conductance regulator) dysfunction.<sup>223</sup> Reduced CFTR function could worsen cystic fibrosis and may contribute to chronic bronchitis onset or progression.<sup>224</sup> ENDS vapor caused mucociliary dysfunction in sheep.<sup>225</sup>

In terms of the development of COPD or lung cancer, animal models will likely serve as the most reliable approach to examine the chronic health effects of ENDS until longer-term studies have been conducted in humans. In vitro studies and animal models suggest that ENDS use increases oxidative stress measures, proteases, inflammation, and DNA damage, all indicators that may increase the risk of COPD and lung cancer.<sup>222</sup> In a study of mice exposed to 12 weeks of ENDS aerosol, sustained extensive DNA damage in lungs, heart, and bladder mucosa was observed.<sup>211</sup> In a longer exposure (54 weeks), animals developed lung adenocarcinomas and bladder urothelial hyperplasia, with lesions being rare in controls.<sup>226</sup> Two studies have shown that long-term e-cigarette exposure in mice induces nicotine-dependent airway changes similar to COPD.<sup>227,228</sup> However, in another study examining long-term exposure to different flavors of nicotine-containing e-cigarette vapor, mice did not develop an emphysema-like syndrome.<sup>209</sup> This suggests that the different chemicals present across ENDS products (tetrahydrocannabinol, flavoring agents, metals) may well have different toxicity profiles. Last, few studies have investigated the effects of e-cigarettes on the pulmonary vasculature. Mice exposed to e-cigarettes had a significant reduction in pulmonary capillary numbers, similar to that seen from conventional tobacco smoke.<sup>228</sup> Ultimately, more studies, both clinical and in animal models, are required to better understand the long-term impact of ENDS product use and the development of cancer.

## EXAMINATION OF E-CIGARETTE AND VAPING PRODUCTS AS CIGARETTE-CESSATION PRODUCTS

Combustible cigarette smoking remains the leading cause of preventable death in the United States, with ≈480 000 deaths per year being directly attributable to smoking-related causes.<sup>207</sup> Given the established risks of cigarette smoking, ENDS products have been evalu-

ated as cessation tools for cigarette use. A recent Cochrane Database of Systematic Reviews performed a meta-analysis on the efficacy of e-cigarettes for smoking cessation and frequency of adverse events related to e-cigarette use.<sup>229</sup> Compared with nicotine replacement therapy (NRT), the analysis found a moderate certainty of evidence that more people quit smoking with nicotine-containing ENDS products compared with NRT (4 studies, 1924 participants; relative effect, 1.53 [95% CI, 1.21–1.93]; [Supplemental Figure 2](#)).<sup>125,229–231</sup> Adverse events were similar.<sup>229</sup> In the Cochrane review, the analysis found that nicotine-containing ENDS products may help more people to stop smoking than no support or behavioral support only (6 studies, 2886 people; relative effect, 2.61 [95% CI, 1.44–4.74]) with low certainty.<sup>232–236</sup> Adverse events were higher at 12 weeks to 6 months in ENDS users compared with those with no support or behavioral support only.<sup>229</sup> One study compared e-cigarettes with varenicline, finding that e-cigarettes were less effective than varenicline (4 of 27 e-cigarette users versus 13 of 27 varenicline users stopped smoking; risk ratio, 0.31 [95% CI, 0.11–0.82]).<sup>229,237</sup> These main study results assessed smoking cessation not complete product cessation. This could mean that participants who quit smoking continued ENDS use.

A recent trial compared nicotine ENDS plus NRT (nicotine patch) with NRT alone and NRT plus nicotine-free ENDS.<sup>238</sup> There was a 7% abstinence rate in the nicotine patch and ENDS treatment groups, which was significantly higher than with the patch alone (2%) and a patch and nicotine-free ENDS (4%).<sup>238</sup> These findings are similar to an early clinical trial examining nicotine-containing ENDS products compared with the nicotine patch alone and nicotine patch plus nicotine-free ENDS condition.<sup>230</sup> Another trial showed that ENDS use may reduce smoking among those given a nicotine-containing ENDS product compared with a nicotine-free ENDS.<sup>239</sup>

Overall, more studies are needed to find the most effective method for cigarette smokers to stop. No current ENDS products have received FDA approval as a tobacco-cessation aid. There is only low to moderate confidence of improved cessation with nicotine-containing ENDS products compared with NRT or behavioral interventions.<sup>229</sup> It is important to note that there are many components of ENDS products, including flavors, device type, device features (eg, power, voltage), and nicotine strength, that, according to experimental work, may be expected to affect cessation efforts.<sup>240</sup> Future trials should make these variables a consideration.<sup>240</sup> Furthermore, because the dose-response relationship between combustible cigarettes and cardiovascular disease is nonlinear, it should be emphasized that complete cessation of the use of all tobacco products should be the ultimate goal.<sup>241</sup>

Although the effects of dual use remain incompletely understood, there are growing concerns that dual use

may make cessation more difficult to achieve, in part by sustaining the delivery of nicotine and supporting addiction. Compared with exclusive users, dual users have a higher prevalence of use of tobacco products and lower harm perceptions for tobacco products.<sup>242</sup> According to a systematic review and meta-analysis that examined the association between e-cigarette use and cigarette smoking cessation among adult cigarette smokers, the odds of quitting cigarettes were 28% lower among dual users compared with users who did not use e-cigarettes.<sup>243</sup> A recent prospective cohort study found that smokers who used ENDS products at baseline were less likely to quit smoking than those who did not use ENDS products at baseline (9.4% stopped smoking versus 18.9%).<sup>244</sup>

From a population health perspective, any potential benefits of ENDS products as effective cigarette-cessation products and the benefits to current smokers stopping cigarettes must be balanced with the risks that nonsmokers start using ENDS products exclusively or as a gateway to further tobacco product use.<sup>245</sup> The negative impact of initiation of ENDS products is a particularly important concern among youth.<sup>246,247</sup> Studies suggest that the rapid uptake of ENDS products among youth coincides with the rapid increases in their unregulated marketing.<sup>248,249</sup> Flavors have been found to be an important motivator for ENDS uptake.<sup>10,250</sup> Exposure to ENDS advertising is associated with lower ENDS harm perceptions and perceived addictiveness<sup>249,251,252</sup> and plays a role in a youth's decision to initiate ENDS use.<sup>253</sup> A recent systematic review suggested an association between exposure to ENDS advertisement and lower harm perceptions about ENDS products, higher intention to use ENDS products, and more ENDS experimentation in adolescents and young adults.<sup>254</sup> ENDS products are not only marketed in traditional media such as television and print advertisements<sup>255</sup> but also aggressively advertised in social media platforms that are highly accessible to youth.<sup>256–261</sup>

## PUBLIC HEALTH EFFORTS, REGULATION, AND MARKETING OF E-CIGARETTES AND VAPING PRODUCTS

### Summary of the Current Regulatory Environment for ENDS Products

The Family Smoking Prevention and Tobacco Control Act, signed by President Barack Obama in 2009, gave the US FDA the authority to regulate the manufacturing, marketing, distribution, and sale of tobacco products.<sup>262–264</sup> The authority, under the FDA Center for Tobacco Products, is different from drug and medical device regulation.<sup>262–264</sup> The FDA cannot ban the sale of certain categories of tobacco products or face-to-face sales at certain categories of retail outlets, cannot require prescriptions for the purchase of tobacco prod-

ucts, and cannot require the reduction of nicotine levels to zero.<sup>262–264</sup> Tobacco product manufacturers are required to submit to the FDA lists of the ingredients and constituents in their products and internal documents on their safety.<sup>262–264</sup> The Family Smoking Prevention and Tobacco Control Act formed the Tobacco Products Scientific Advisory Committee, which reviews data on the safety, dependence, and health issues related to tobacco products.<sup>265</sup> The committee then provides advice, information, and recommendations to the FDA such as whether a tobacco product should qualify as modified risk because it reduces harm or reduces exposure to an ingredient compared with another pre-existing product.<sup>266</sup> Of note, the initial Family Smoking Prevention and Tobacco Control Act did not address ENDS products because there were few in the market at that time.

In 2016, the FDA finalized a rule that deemed ENDS and all other products that meet the definition of tobacco product to be subject to the agency's authority under the Federal Food, Drug and Cosmetic Act.<sup>267</sup> Besides prohibiting the sale of e-cigarettes to youth <18 years of age, this new rule prohibited false and misleading marketing claims and required premarket review. In January 2020, the FDA issued an enforcement policy to prohibit sales of certain flavored e-cigarettes. This policy, however, applies only to cartridge- or pod-based e-cigarette products, defined as "any small, enclosed unit (sealed or unsealed) designed to fit within or operate as part of an electronic nicotine delivery system," and exempted all tobacco or menthol-flavored products.<sup>268</sup> The policy also exempted self-contained, disposable products and e-liquids for open-tank or refillable ENDS products. In response, many of these products such as the disposable e-cigarette Puff Bar with synthetic nicotine<sup>269</sup> gained dramatic market share after this FDA policy took effect.<sup>270</sup> Separately, the FDA proposed banning menthol cigarettes and all flavored cigars in April 2022.<sup>271</sup> To minimize adolescents and youth tobacco initiation, in December 2019, the president signed legislation to raise the federal minimum legal age of sale of tobacco products, including ENDS products, to 21 years, known as Tobacco 21, effective immediately.<sup>272</sup>

Despite all these regulatory approaches, there are continued gaps in regulation that allow the tobacco industry to continue to sell many tobacco products that are easily accessible to children and young adults.<sup>273</sup> The original Family Smoking Prevention and Tobacco Control Act was limited to 4 tobacco products (smokeless tobacco, roll-your-own tobacco, cigarettes, cigarette tobacco); cigars, e-cigarettes, pipe tobacco, and several other tobacco products were included later.<sup>274</sup> The FDA also has to contend with constant innovation in the tobacco product marketplace such as the emergence of pod-based vaping products in 2015, which led to huge increases in the rates of ENDS use among youth. Puff Bar and

some other e-cigarettes and oral nicotine products were also able to evade FDA flavor restrictions and premarket review requirements by claiming to be tobacco-free synthetic nicotine,<sup>275</sup> which the FDA did not have authority to regulate as a tobacco product until new legislation was enacted in March 2022; that legislation gave the FDA the authority to regulate all tobacco products regardless of the nicotine source.<sup>276</sup> Future innovations in the tobacco product marketplace continue to pose challenges to the FDA's regulatory authority.

### Current Public Health Efforts to Reduce E-Cigarettes and Vaping Products in Youth and Young Adults

As indicated previously, the rates of ENDS use are high among adolescents and young adults.<sup>14</sup> This is particularly true of youths from underresourced populations: those with lower socioeconomic status,<sup>277</sup> those who identify as LGBTQ (lesbian, gay, bisexual, transgender, and queer), individuals from lower educational backgrounds, those from underserved communities, and those with mental health conditions.<sup>278,279</sup> Children are initially attracted to these products for their flavors,<sup>280,281</sup>; many then move on to regular use and nicotine addiction and then may transition to using other tobacco products such as combustible cigarettes.<sup>282–285</sup> Furthermore, ENDS use positively correlates with other substance use and abuse, including alcohol and marijuana.<sup>286</sup> The alarming prevalence of ENDS use among youth and the unknown potential for cardiovascular harm and toxicity emphasize the critical need for public health efforts to prevent and reduce the use of ENDS products among youth.

In response, several public health efforts aimed at preventing e-cigarette and vaping product use among youth and young adults have been developed. These include school-based interventions, community-based interventions, and public education intervention (as outlined in a review by the Ontario Tobacco Research Unit).<sup>287</sup> It is important to note that many of these interventions have not yet been empirically tested but have been implemented in various settings.

School-based prevention interventions include the CATCH my Breath Youth E-Cigarette Program,<sup>288</sup> the Tobacco Prevention Toolkit by Stanford University,<sup>289</sup> and "E-Cigarettes: What You Need to Know (Teacher's Guide)" by Scholastic.<sup>290</sup> The majority of these curriculums are aimed at middle and high school-aged students, and their main mode of delivery is teacher/educator-delivered educational materials. In a pilot in Texas middle schools, those receiving the CATCH my Breath program (compared with students in control schools) were observed to have lower prevalence of e-cigarette use after 16 months.<sup>288</sup> Other programs that are being developed and tested for use among students include smoke screen, an e-cigarette video game, and ASPIRE, a learn-

ing module aimed at middle and high school students, prior iterations of which demonstrated effectiveness at preventing cigarette use.<sup>291,292</sup> Several community-based text message programs have been created by Truth Initiative: This is Quitting, BecomeAnEX, and others are aimed at both cigarette and e-cigarette users.<sup>293,294</sup> This is Quitting was designed with youth and young adults in mind, whereas the other 2 programs are aimed at adult tobacco users.<sup>293,294</sup> Preliminary evidence on This is Quitting among young adults suggests that the intervention doubled the self-reported 30-day abstinence rates.<sup>294</sup>

Public education campaigns that use various approaches, including paid ads, social media, and materials such as fact sheets and posters, have also been developed. One example is the US FDA's Real Cost E-Cigarette Prevention Campaign, focused on debunking false information about e-cigarettes through advertisements, which has been shown to increase negative views of vaping among adolescents and to lower their intentions to use e-cigarettes.<sup>295</sup>

In summary, few empirically tested prevention and cessation programs for youth ENDS use exist. Some strategies that could deter minor ENDS use and improve adolescent ENDS cessation include the use of novel technology (ie, text messages, social media)<sup>296</sup> that has been used extensively to advertise ENDS products to youth,<sup>259–261</sup> educational efforts focused on parents and health educators, focused assessment and treatment by pediatricians and other health care professionals, and use of other behavioral methods such as incentives and cognitive behavior therapy that have been successfully shown to promote smoking cessation among youth.<sup>297–299</sup> Future work on developing and testing effective interventions is critically needed, and such work is being supported by the End Nicotine Addiction in Children and Teens initiative funded by the American Heart Association in 2020. This critical initiative funds scientific research at Boston University, The Ohio University, and Yale University to understand the health impact of ENDS use on youth and young adults and to develop interventions.<sup>300</sup> Furthermore, a recent presidential advisory from the American Heart Association delivered several positions on newer tobacco product regulation (Table 2).<sup>301</sup>

### KNOWLEDGE GAPS AND RESEARCH PRIORITIES

ENDS products have undeniably been increasing in popularity, particularly among young adults and teens, in the past decade. The constituents of these products often include nicotine, which is well established to have negative health effects and strong addictive properties. Other ingredients, particularly in flavored products, have known health risks. Because ENDS products are not regulated as classic therapeutic drugs or devices, there are no

**Table 2. AHA Positions on Newer Tobacco Product Regulation**

| Issue                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tobacco endgame                   | The AHA supports ending the use of all combustible tobacco products while ensuring that other products do not addict the next generation of youth and adolescents and achieving a realistic goal of getting to ≤5% tobacco use prevalence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Underresourced populations        | Tobacco control and prevention efforts and regulation should be targeted and tailored to at-risk populations, including youth and adolescents, those who live in rural areas, racial and ethnic groups with high tobacco use, those with mental health conditions, individuals with less education and low income, and those who identify as LGBTQ.                                                                                                                                                                                                                                                                                                                                                                                                    |
| Nicotine reduction strategy       | The AHA favors lowering nicotine content in all combustible tobacco products. The AHA supports lowering nicotine concentrations in all combustible tobacco products to reduce tobacco-related mortality.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Flavorings                        | The removal of all characterizing flavors from all tobacco products is essential for reducing their appeal to youth.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| Market review                     | The AHA supports restricting the marketing of JUUL and other similar e-cigarettes until their health risks to youth and adolescent users are clearly assessed and their potential benefits and harms in promoting tobacco cessation among adults are better understood. The agency should suspend internet sales of these products until adequate mechanisms and rules for age verification are established. In addition, the ban on underage sales by retailers should be effectively enforced, and the FDA should require that these products be submitted for review.                                                                                                                                                                               |
| Marketing and advertising         | The AHA supports robust FDA regulation restricting all tobacco marketing and advertising to youth and vulnerable populations, including the use of television, radio, and print ads and commercials; celebrity endorsement; movie placements; price promotions; free sampling; branded events; and nontobacco merchandise.                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Public education                  | There is a significant need for robust public education about these newer tobacco products and the harms they pose, especially for youth and adolescents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Tobacco 21                        | The AHA advocates for Tobacco 21 laws that incorporate all tobacco products to minimize youth and adolescent initiation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Comprehensive smoke-free air      | Smoke-free laws should explicitly include aerosolized, alternative nicotine delivery systems, and combustible products in comprehensive smoke-free air laws to ensure that there is no passive exposure to any harmful constituent byproducts or risk of renormalizing tobacco use.                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| Health care professional guidance | Health care professionals should screen for all tobacco product use and counsel cessation. Young patients should be screened for newer tobacco use and substance abuse and counseled on the dangers of these products. A previous AHA policy statement elucidated how clinicians should advise adult patients about cessation. Youth substance use prevention programs should target reduction of e-cigarette and cigar use. In the intersection between the health care system and public health, there is a need to develop public health messages that accurately convey the scientific data on the potential harm of newer tobacco products and differentiate the absolute from relative harm of these products compared with combustible tobacco. |

AHA indicates American Heart Association; e-cigarette, electronic cigarette; FDA, US Food and Drug Administration; and LGBTQ, lesbian, gay, bisexual, trans-gender, and queer.

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dedicated long-term safety studies. Critical questions remain unanswered about the short-term and, in particular, long-term health effects of ENDS products. Because the products have only recently gained widespread use, decades of prospective or retrospective data are not yet available to examine the long-term health effects of cigarettes. Early analysis suggests some utility of ENDS as a smoking cessation product; however, any benefit needs to be juxtaposed with a clear understanding of the health risks of the ENDS products themselves and the risks of product availability leading to nonsmokers initiating ENDS use. Regulatory bodies and governments worldwide will be required to examine these aspects critically as ENDS manufacturers market and distribute these products. The advent of EVALI emphasizes the pressing need to better understand the role of constituents used in these products. The large informal market, including online distribution, homemade vaping liquids, and locally manufactured vaping liquids, creates an ENDS user base that is difficult to regulate and study. Although more studies are desperately needed to understand the full health risk profile of different types of ENDS devices and ENDS liquids, claims that ENDS products present absolutely no health risks are false according to the limited, but growing, evidence available. Future research should focus on understanding the short- and long-term cardiopulmonary impact of ENDS product use and commonly used constituents such as flavoring agents in both human and animal exposure models. Studies on patients with preexisting cardiopulmonary disease (coronary artery disease, COPD) comparing ENDS products users with both smokers and nonsmokers will be important. Long-term effects of ENDS use will eventually be determined epidemiologically. However, to accurately balance the risks and benefits of ENDS on a population scale, long-term animal exposure models will be required in the interim before decades-long evidence emerges. Policies on the use of ENDS products as harm reduction tools need to be balanced with the fact that ENDS use is rising rapidly in young adults and youth who may have never started using combustible tobacco products.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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| Writing group member    | Employment                                                                  | Research grant                                                                                                                                                                                               | Other research support | Speakers' bureau/honoraria | Expert witness                  | Ownership interest                                                                              | Consultant/advisory board                                                 | Other                                                                      |
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| Naomi M. Hamburg        | Boston University School of Medicine                                        | None                                                                                                                                                                                                         | None                   | None                       | None                            | None                                                                                            | None                                                                      | None                                                                       |
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\*Modest.

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## Reviewer Disclosures

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